Mother-Infant Separation in Group-Living Rhesus Macaques: A Hormonal Analysis

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Adrenocortical responsiveness to separation in group-living rhesus macaque mother-infant pairs was determined. Cortisol evaluations were complemented by behavioral observations of mother-infant interactions and group social behavior. Infants, but not their mothers, showed cortisol elevations that could be attributed specifically to separation. Steroid-behavior relationships suggested that the infants of high dominant mothers evinced the greatest adrenocortical response to separation. These data exemplify the complexities involved in determining endocrine/behavior relationships in a group-living situation.

The effects of maternal separation on the behavior of infant rhesus monkeys have been well studied by numerous investigators. Although significant individual variation is common, infants typically display marked protest upon separation from their mothers. If separation is continued beyond several days the infants often enter a "despair" stage (Kaufman & Rosenblum, 1967; Spencer-Booth & Hinde, 1971). Behavioral disruption is also common upon reunion of mother and infant (Hinde & Spencer-Booth, 1971; Rosenblum & Youngstein, 1974). Individual variation in the postseparation response has been attributed primarily to differences in the quality of the mother-infant relationship prior to separation and to the social tension experienced by the mother (Hinde & McGinnis, 1977). In general, the extent of the behavioral disruption due to separation has been evaluated as an index of the infant's attachment to its mother (Hinde & Spencer-Booth, 1971). Recently, the effects of separation on adrenocortical responsiveness as well as on behavior has been studied. A study of mother-reared and surrogate-reared infant squirrel monkeys has revealed significant separation-induced elevations in plasma cortisol of both mother and infant (Mendoza, Smotherman, Miner, Kaplan, & Levine, 1978).

The current study sought to record separation-induced effects on adrenocortical activity as well as to monitor ongoing behavior in group-living rhesus monkeys.

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Specifically, we hoped to evaluate possible relationships between the cortisol response to separation and characteristics of the mother-infant interaction and group social structure. Many different behavioral measures were obtained; however, this paper will treat only those behaviors that were found to be correlated with steroids.

Methods

Subjects

Five mother-infant rhesus monkey pairs (*Macaca mulatta*) were the subjects of our investigation. Four of the infants were male and 1 was female. During the course of the study, the infants' ages ranged from 6 to 20 weeks. The monkeys lived in a social group consisting of 2 adult males, 2 adult nonlactating females, a mother with an older infant, and the 5 experimental mother-infant pairs. The mothers ranged in age from 5 to 6½ years. The animals were housed in a large outdoor cage (5 x 10 x 3 m) which adjoined a smaller enclosed room (3 x 5 x 3 m). They had lived in this setting for 1½ years before the start of this study. A second colony of rhesus macaques was within view of the experimental group.

The investigation was conducted from June until September. All lighting was natural and food and water were available ad lib.

Separations

All separations and observations were conducted between 0800 and 1200 hours. Once each week 2 of the mother-infant pairs were captured by the 1st author and a caretaker familiar to the animals and removed from the group. This procedure generally took 2-5 min. Pairs were captured in a randomized order. They were then housed in standard primate cages either individually or in pairs for a period of 3 hr. Individual housing was designated the "separation" condition and paired housing the "control" condition. Using this design we were able to distinguish the adrenocortical response to separation apart from the effects of capture and removal from the group. During separation the mothers and infants were kept adjacent to the large outdoor cage. They were permitted auditory and visual contact both with each other and with the remaining members of the group.

Over the course of the investigation each pair experienced the separation and the control condition 3 separate times. Conditions were ordered so that no animal was exposed to the same experimental procedure during 2 consecutive weeks.

Blood Sampling and Plasma Cortisol Determination

At the end of 3 hr, the animals were removed from the holding cages in a random order and .5 ml of blood was quickly collected from the saphenous vein (within 2 min) in heparinized syringes. Unfortunately, the animals could not be isolated rapidly enough from the large group cage in order to determine resting levels of plasma cortisol.

The blood samples were centrifuged at 2000 rpm and the plasma was removed and

quick-frozen for subsequent determination of cortisol levels by a radioimmunoassay technique. Plasma cortisol was assayed using antiserum No. F21-53 (Endocrine Sciences, Tarzana, California) according to the method described previously by Klemm and Gupta (1975). An aliquot of plasma was extracted with ethylacetate prior to assay. A small amount of titrated hydrocortisone was then added to each plasma sample. The plasma-trace mixture was extracted twice with ethyl acetate (40:1) after equilibrating at room temperature for 30 min. The two organic phases were pooled and blown dry under N_2 at 37°C. The dried extracts were covered and stored at −20°C until assay. For the assay, the extracts were brought back to room temperature. Borate buffer was added to each extraction tube and duplicate aliquots were assayed. An aliquot was also counted for recovery and corrections for recovery were made on each sample. The heat step and methanol were omitted from the assay procedure as the plasma had been extracted. Otherwise the assay was performed as previously described.

Observations

Behavioral data were collected 3 times per week on days when separations were not conducted. The animals were given 4 weeks to adjust to the observers before separations were begun. Further, the observers were never present during the separation or blood sampling procedures in order to minimize the effect of the observers while monitoring behavior. During observations, each rhesus pair was followed for 30 consecutive minutes; the order in which the pairs were followed varied systematically.

Mother-infant interactions and group social behavior were recorded by separate observers who sat near the perimeter of the outdoor cage. Data were collected on check sheets coded for specific behaviors, using 30-sec time divisions. Mother-infant interactions which we defined as events were recorded each time they occurred. These behaviors included making or breaking ventral contact, approaching or leaving one another, vocalizations, and maternal rejections, that is, preventing the infant from reaching the ventral body surface. Behavioral states were recorded by a "1"-"0" sampling method or Hansen frequency (Altmann, 1974). The behaviors recorded using the Hansen frequency were mother-infant contact and proximity. The group social behaviors were also divided between events and states. Approaches, leaves, huddling, play bouts, threats, avoidance behavior, and sexual behavior were considered events and were noted each time they occurred regardless of whether or not they involved the focal dyad. Grooming and proximity of mother and infant to other group members were considered behavioral states and were recorded as either occurring or not occurring during each 30-sec time interval.

Data Analyses

All behavioral measures were tabulated and the mothers and infants were ordered in terms of the amount of time they spent engaged in each behavior. Similar orderings were performed for the mean plasma cortisol value of each animal in the separation and the control conditions. Spearman rank order correlation coefficients were then calculated between each behavior and the 2 steroid measures for both mothers and infants.

In addition, mother and infant cortisol values were analyzed separately using a 2-factor, repeated measures analysis of variance.

Results

After plotting the cortisol values as a function of the age of the infants, we did not see any consistent change over weeks in the steroid response of either the infants or the mothers. From 10 to 20 weeks of age, the infants' median plasma cortisol levels in both separation and control conditions did not vary in any predictable manner. The mothers' values remained virtually constant in both experimental conditions (Fig. 1).

Because only 1 female was among the infants studied we can draw no conclusions based on the gender of the infant. (However, neither the behavioral nor the endocrine measures for the female varied far from mean values.)

An overall analysis of variance of the steroid values revealed considerable individual variation. Both the 5 infants ($F = 2.86$, $df = 4/20$, $p < .05$) and their mothers ($F = 4.27$, $df = 4/20$, $p < .01$) displayed significant variation in measures of plasma cortisol (Fig. 2).

Analysis of cortisol levels between the separation and control conditions revealed a significant difference only in the infants' corticoids ($F = 12.51$, $df = 1/20$, $p < .01$; Fig. 2). The mean infant value for the control condition was 28.07 (± 2.20) μg/100 ml whereas the mean separation value rose to 39.31 (± 2.59) μg/100 ml. This differential steroid response to the 2 conditions was not found in the mothers ($F = .03$, $df = 1/20$, $p > .25$). The control cortisol mean for the mothers was 35.15 (± 1.58) μg/100 ml; the separation value was 35.45 (± 1.52) μg/100 ml.

In light of the significant individual variation in plasma cortisol values for both mothers and infants, we attempted to relate these values to measures obtained for mother-infant interactions and group social behavior. We had anticipated that the Spearman rank order correlation coefficients would reveal interesting relationships between endocrine and behavioral measures. However, only the correlation of infant steroid levels with maternal dominance status reached statistical significance. The more dominant the mother, the higher the cortisol levels of her infant in both the separation

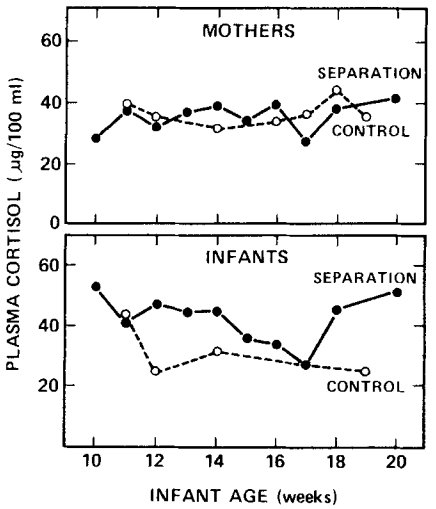


Fig. 1. Plasma cortisol expressed as a function of infant age. Entries represent median values.

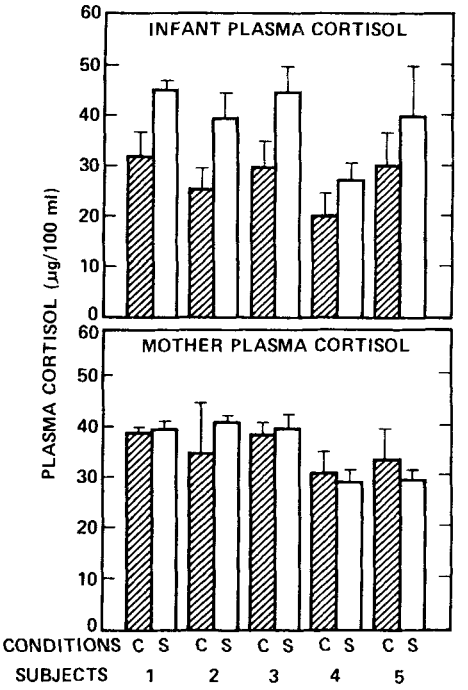


Fig. 2. Plasma cortisol for individual subjects in control and separation experimental conditions. Entries represent mean values (± S.E.M.).

and the control condition (Dominance was determined by approach/avoidance behavior. Low rank mothers ranked highest on avoidances of others and lowest on avoidance of them by others. High ranked mothers ranked lowest on avoiding others and highest on being avoided by others. Mothers that were avoided the most also threatened the most and mothers that avoided the most were threatened the most. Other investigators have reported the approach-avoidance measure as a consistently reliable indicator of rhesus status [Bernstein, 1970; Rowell, 1972].)

Discussion

The present investigation focused upon the effects of maternal separation on adrenocortical responsiveness of group-living mother and infant rhesus macaques. We hoped to discover which features of mother-infant interaction and group social behavior were related to cortisol levels and which were independent of the adrenocortical response.

Inspecting the steroid data in terms of infant age, we found virtually no change from Week 10 through Week 20 in either the mothers or the infants. These data parallel those reported by Hill, McCormack, and Mason (1973). Behaviorally, however, these weeks are typically a time of continual change in the mother-infant relationship (Hinde & Spencer-Booth, 1967; Jolly, 1972). Due to increasing independence of the infant and rejections by the mother, we had thought that we might also see a

developmental progression in steroid levels, although this was not the case. Note that our animals (as well as those in Hill *et al.*) experienced repeated stressors; the weekly disturbance in the home cage may have altered a normal course of development in adrenocortical responsiveness.

Only the infants showed a clear and specific steroid response to maternal separation. This finding parallels the reaction of group-housed squirrel monkeys to maternal separation (Coe, Mendoza, Smotherman, & Levine, 1978).

The mothers' plasma cortisol levels were essentially identical for the 2 procedures. As we have no baseline data for these animals, we can draw only tentative conclusions. Considering base values reported by other investigators (Feldman & Brown, 1976), we guess that the mothers exhibited elevated corticoids in both the separation and the control condition. Apparently capture and removal from the group was, in itself, a severe enough stress for the mothers that subsequent removal of their infants caused no discernible increase in adrenocortical activity. Unlike the infants, the mothers' steroid response contrasts with the response of squirrel monkeys reported by Coe *et al.* (1978). The squirrel monkey mothers showed a greater cortisol rise when separated from their infants for 30 min than when captured, separated momentarily, and returned to the group. The discrepancy in response patterns for rhesus and squirrel monkey mothers may reflect a true species difference or may be simply the result of the procedural differences in the 2 studies.

The result of special significance is the close correlation between infant steroid levels and maternal dominance status. The infants of highly dominant mothers displayed a greater adrenocortical response in both separation and control conditions than did infants of low ranking mothers. We found no correlation between maternal dominance status and maternal steroid values. Other investigators have reported an inverse relationship between dominance and the activity of the pituitary-adrenal axis in adult rhesus monkeys. However, this finding existed only when dominance was also inversely related to social stress (Chamove & Bowman, 1976; Sassenrath, 1970). In the present study the most dominant females in the group appeared not to be, in fact, the least stressed animals. They frequently appeared more agitated than did the middle ranked animals.

The correlation between social rank and infant plasma cortisol values highlights the complexities involved in the mother-infant relationship in group-living rhesus monkeys. The mother's status within the group affects both her behavior toward others and their reaction to her. In turn, these social conditions must have ramifications upon the mother-infant interaction whether the mother is comfortably stable in her dominance position, i.e., without threat from others, tensely maintaining a high rank or actively avoiding the aggression of others. The nature of the mother-infant interaction is then reflected in the infant's response to separation. In the present study, this multiple interaction is exemplified by the differing endocrine responses for infants of high and low ranking mothers.

Notes

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